

Indian Pines Hyperspectral Image Visualization

Visualizing, exploring, and dimensionality-reducing a 200-band AVIRIS hyperspectral scene using PCA and t-SNE.

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Overview

The Indian Pines scene is a benchmark hyperspectral dataset captured by the AVIRIS airborne sensor over a mixed agricultural and forested area in Indiana. The image is 145 by 145 pixels, each pixel described by 200 spectral bands, and annotated with 16 land-cover class labels. This project builds a complete visualization and dimensionality-reduction pipeline for the scene, from raw data loading through class-level spectral analysis and low-dimensional embedding.

Goals

Produce clear, publication-quality visualizations of the hyperspectral cube and demonstrate that PCA and t-SNE can meaningfully separate the 16 land-cover classes in a reduced two-dimensional space.

Objectives

- Load and inspect the full hyperspectral data cube and corresponding ground-truth label map.
- Compose a false-color RGB image by selecting three representative spectral bands.
- Visualize the class distribution and overlay the ground-truth mask on the false-color image with a class legend.
- Plot mean spectral signatures for each class by sampling 1 percent of pixels per class.
- Apply PCA and t-SNE to the cube, visualize 2D scatter plots colored by class, and reconstruct scene images from the leading components.

Approach

The raw hyperspectral cube is reshaped from a 145 by 145 by 200 volume into a 2D pixel-by-band matrix for analysis. PCA is applied directly to this matrix; t-SNE is run on the first 30 PCA dimensions to keep computation tractable over roughly 21,000 pixels. Scatter plots of the first two components are colored by land-cover label to assess class separability. Scene images are reconstructed by reshaping the reduced representations back to 145 by 145.

Outcomes

- PCA's first two components capture approximately 92 percent of total spectral variance (PC1 approximately 68 percent, PC2 approximately 24 percent).
- False-color composites and ground-truth overlays clearly distinguish the 16 land-cover classes visually.
- Mean spectral signatures reveal distinct reflectance profiles across vegetation, crop, and built-up classes.
- PCA and t-SNE scatter plots demonstrate meaningful class clustering in two dimensions, confirming that the high-dimensional spectral data has strong low-dimensional structure.